This section provides knowledge and understanding of photons, the photoelectric effect, de Broglie waves and wave–particle duality.

The topic of quantum physics provides an ideal opportunity for learners to carry out Internet research and present their findings as short PowerPoint presentations (HSW3). Learners have the opportunity to appreciate how scientific theory of quantum physics developed over time (HSW1, 2 and 7).

4.5.1 Photons

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the particulate nature (photon model) of electromagnetic radiation	
(b)	photon as a quantum of energy of electromagnetic radiation	
(c)	energy of a photon; $E = hf$ and $E = \frac{hc}{\lambda}$	
(d)	the electronvolt (eV) as a unit of energy	
(e)	**(i)** using LEDs and the equation $eV = \frac{hc}{\lambda}$ to estimate the value of Planck constant h	No knowledge of semiconductor theory is required.
	(ii) Determine the Planck constant using different coloured LEDs.	**PAG6**